

L15-4522
EE-2A

EE227 – Digital Logic Design
Midterm-I

Date: September 19, 2016
Total Marks: 45

Time Allowed: 60 mins
Weightage: 15

INSTRUCTIONS: Show complete working for each question, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Attempt each question only once. **Calculators are not allowed.**

Q#1:

[8+8 Marks]

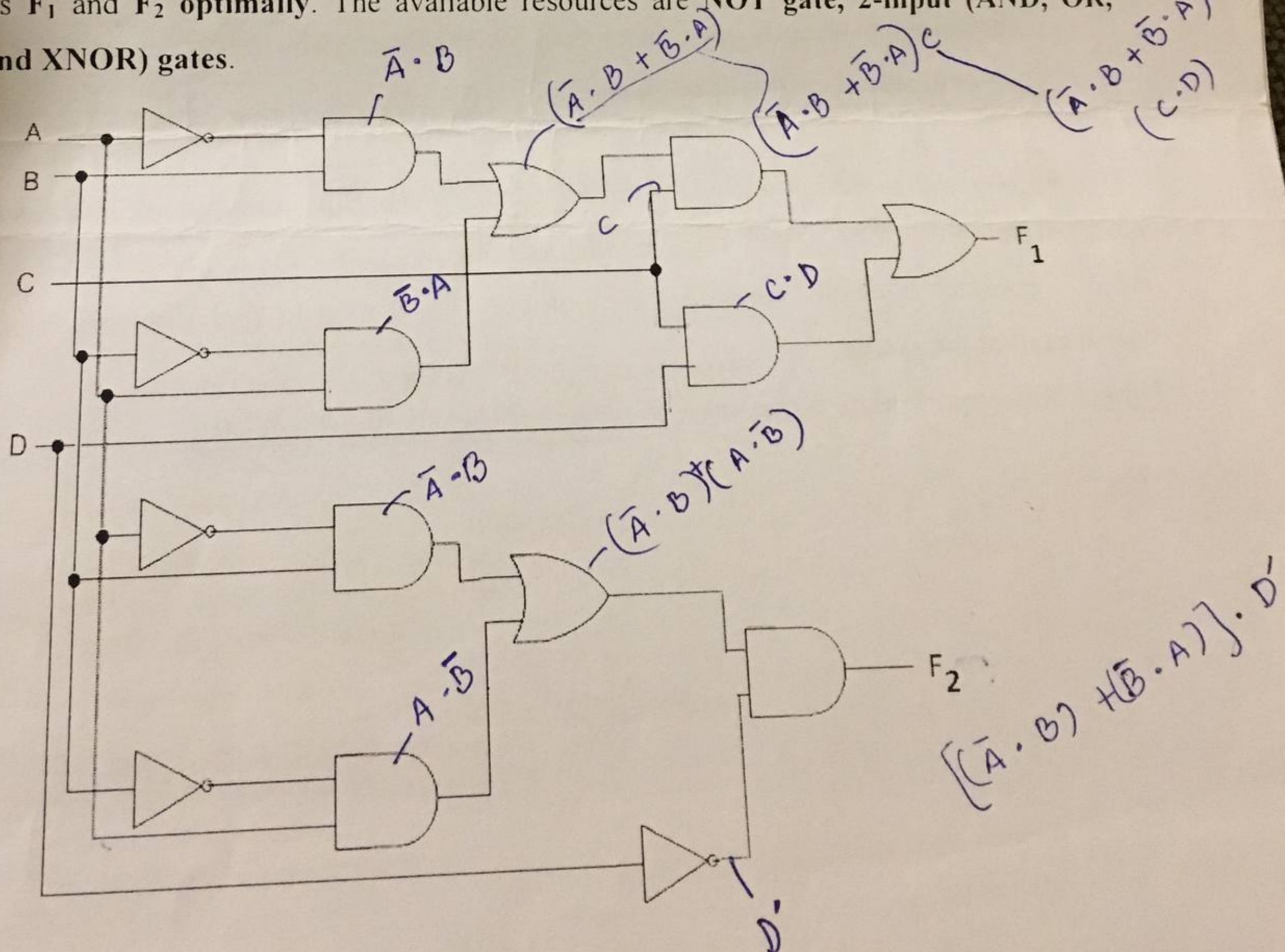
- a) $(167)_8 + (2EA)_{16} = (\quad)_5$
b) Use algebraic manipulation to reduce the following Boolean expression,

$$F(w, x, y, z) = [yz + w\bar{x}y(\bar{w} + z)] + \overline{[x\bar{y} + (x + y) + \bar{z}]} y$$

Q#2:

[8 Marks]

In the figure given below, logic circuit for Boolean functions F_1 and F_2 are designed using the **NOT** and **2-input (AND, OR)** gates. You are required to redesign the logic circuit for Boolean functions F_1 and F_2 **optimally**. The available resources are **NOT** gate, 2-input (AND, OR, XOR, and XNOR) gates.



Q#3:

Find minimal **SOP** expression for the Boolean function **F** given below using **K-map**: [8 Marks]

$$F(A, B, C, D) = \sum m(0, 1, 2, 4, 7, 8, 9, 10, 12, 15)$$

Q#4:

[13 Marks]

KHAADI fashion store, DHA branch Lahore, is planning to celebrate its upcoming 20th birthday by offering **special discount cards** and **gift coupons** to its customers in October, 2016. There will be two discount cards (silver and golden). The customers will be given discount cards under certain conditions. The customer who does not fulfill condition for any one of the two cards then random gift coupon will be given to the customer. The details are as under:

- If the customer possesses old KHAADI discount card that is not expired and the customer has recently spent more than 50,000 rupees while shopping at KHADDI then **golden discount card** will be given.
- If the customer possesses old KHAADI discount card but it has been expired and the customer has recently spent more than 50,000 rupees while shopping at KHAADI then **silver discount card** will be given.

You are required to design a digital system indicating what discount/gift offer a customer has been given. Show all necessary steps to reach optimal design for the required system, from truth table to minimal Boolean expressions for outputs. Use K-map to find Boolean expression for each output of the system.

Hint: Choose appropriate binary code to represent each discount/gift offer.



Course: Digital Logic Design
Program: BS(Electrical Engineering)
Duration: 60 Minutes
Paper Date: 14-Nov-16
Section: ALL
Exam: Midterm-II

Course Code: EE227
Semester: Fall 2016
Total Marks: 45
Weight: 15%
Page(s): 1
Roll No:

Instruction/Notes:

LIS-4522

Show complete working for each question, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Attempt each question only once. Calculators are not allowed.

Question#1: a) Implement the Boolean function $F_1(A, B, C, D) = \sum m(0, 1, 2, 3, 4, 5, 9, 10, 11, 12, 13, 15)$ using 4x16 decoder with active low outputs and minimum extra logic. Available resources are 4x16 decoders, NOT gate, 2-input (AND, OR, NAND, NOR, XOR) gates. [7 marks]

b) Perform the arithmetic operations in binary for the given decimal numbers. Use 2's complement representation for negative numbers. Indicate whether overflow occurs or not for each arithmetic operation. Consider each number to be 6-bit wide. [8 marks]

- i. $-23 - (-3)$
- ii. $-23 - 17$

Question#2: Design a combinational circuit that takes 3-bit pattern as input and outputs binary code of bit position of the first '1' in the pattern reading from MSB (2^{nd} position) to LSB (0^{th} position). An additional output variable **V** is required along with binary code to indicate that the binary code is valid or not i.e., if the input pattern is '000' then the output **V** should be '0' to indicate that the binary code is not indicating the bit position of first '1' and we don't care about the binary code if $V = 0$. Design the required circuit using dual 4x1 MUXs and minimum additional logic. Available resources along with dual 4x1 MUXs are NOT gates, 2-input (AND, OR, NAND, NOR) gates. [15 marks]

Question#3: Design an optimized digital circuit to perform the arithmetic operation $3x + 3c$, where $c = 2$. Here, x is unsigned 4-bit number. Available resources are 3-bit Adders, Half Adders (HA), Full Adders (FA), Full Subtractors (FS), 4x16 decoders (with active high outputs), 8x3 priority encoder, NOT gates, 2-input (AND, OR, NAND, NOR, XOR, XNOR) gates. Show complete working. [15 marks]